

QuantFusion

Development Roadmap

Phase-by-phase build plan for the QuantFusion AI Quant Trading System. Each phase produces a working deliverable that can be tested before moving to the next.

Total Phases	7 phases
Estimated Timeline	12-16 weeks to live trading
Approach	Build one agent at a time, test, validate, then add the next
First Live Trade	End of Phase 4 (paper trading from Phase 3)
ML Integration	Phase 6 (after collecting enough trade data)
Full Autonomous	Phase 7 (all agents live, ML optimised)

Timeline Overview

Each phase builds on the previous one. Nothing gets skipped. Every phase has a clear deliverable and a milestone that must be hit before moving forward.

Phase	Name	Duration	Key Deliverable	Agent(s) Built
1	Foundation	Week 1-2	Local dev environment + data pipeline running	Data Agent
2	The Brain	Week 2-4	All indicators calculating live on real data	Math Agent
3	Paper Trading	Week 4-6	System taking paper trades with full checklist	Position + Execution
4	Risk Cage	Week 6-8	All risk controls active, paper trading validated	Risk Agent
5	Context Layer	Week 8-11	LLM reading news and adjusting scores	Research Agent
6	Learning Loop	Week 11-14	ML models trained, weights optimised	Analytics Agent + ML
7	Full Deploy	Week 14-16	Live trading with real capital, all agents active	Full system

Phase 1: Foundation — Data Agent

Week 1-2

Before anything can work, data must flow. This phase sets up the development environment and builds the Data Agent — the plumbing that feeds everything else.

Week 1: Environment setup

Task	Details	Tools
Python environment	Set up virtual env, install core libraries	Python 3.11+, pip, venv
Install ccxt	Unified exchange API library	pip install ccxt
Install pandas-ta	Technical indicator library	pip install pandas-ta
Install asyncio/websockets	Real-time data streaming	pip install websockets aiohttp
Set up PostgreSQL	Local database for historical data	PostgreSQL + psycopg2
Set up Redis	In-memory cache for Ontology state	Redis + redis-py
MEXC API keys	Create API key on MEXC (read-only first)	MEXC account settings
Bybit API keys	Backup exchange API key	Bybit account settings
Project structure	Create folder structure for all agents	FastAPI project scaffold

Week 2: Data Agent live

Task	Details
WebSocket price stream	Connect to MEXC WebSocket, receive BTC/ETH/SOL/XRP ticks in real-time
OHLCV candle builder	Aggregate ticks into 5M, 15M, 1H candles locally
Funding rate polling	REST API call every 8 hours, store in database
Open interest polling	REST API call every 1 minute, store in database
Volume tracking	Calculate current volume and MA(5), MA(10), MA(20)
Historical data download	Download 2 years of OHLCV data for all 4 pairs, all 3 timeframes
Data validation	Detect stale data (no update > 10 sec), missing candles, corrupt values
Failover test	Simulate MEXC disconnect, verify switch to Bybit within 2 seconds
Ontology state	Write all live data to Redis shared state — first version of the Ontology

MILESTONE Data Agent streaming live BTC/ETH/SOL/XRP data with failover working

Phase 2: The Brain — Math Agent

Week 2-4

Now that data is flowing, the Math Agent calculates all indicators in real-time and implements the confirmation checklist. This is the quantitative core of the system.

Week 3: Indicator engine

Task	Details
RSI calculation	Implement dual RSI (6 and 24 period) across all 3 timeframes for all 4 pairs
MACD calculation	MACD line, signal line, histogram (12, 26, 9) — detect shrinking/growing bars
EMA 200	200-period EMA on all timeframes — price above or below check
Bollinger Bands	20-period, 2 std dev — upper, middle, lower band positions
ATR	14-period ATR for stop loss sizing — outputs dollar value for each pair
Volume ratio	Current candle volume vs MA(20) — flag when > 1.5x and > 3x
Correlation matrix	Rolling 30-day correlation between all 4 pairs — update every hour
Indicator validation	Compare calculated values against MEXC chart values — must match

Week 4: Checklist + trigger + backtesting

Task	Details
Trigger gate	RSI threshold detection — fires when RSI crosses below 25/28/30 or above 70/72/75
5-point checklist	When trigger fires: check EMA trend, MACD momentum, volume, funding, OI
Score calculation	Sum confirmations — output 0 to 5 score
Session detection	Determine current session (Asia/London/NY) based on UTC time
Session thresholds	Apply different minimum scores per session
Higher TF override	If 1H is bearish, block 5M/15M long signals
Backtest engine	Run the full checklist against 2 years of historical data
Backtest results	Calculate win rate, expectancy, Sharpe, profit factor on historical signals

MILESTONE Backtest shows positive expectancy — the edge is statistically validated

Phase 3: Paper Trading — Position + Execution Agents

Week 4-6

The system starts taking simulated trades. No real money yet. Position Agent sizes the trades, Execution Agent places paper orders. This validates the strategy works in real-time, not just in backtests.

Task	Details
Position sizing formula	Implement: $\text{risk} = \text{balance} \times 0.25\%$, $\text{stop} = 1.5 \times \text{ATR}$, $\text{size} = \text{risk} / \text{stop}$
DCA layer logic	Layer 1 at threshold, Layer 2 at threshold-5, Layer 3 at extreme — each with own checklist
Portfolio exposure tracking	Track total open exposure, enforce max 3 concurrent positions
Paper order engine	Simulated order placement — tracks entry, stop, TP without real exchange orders
Paper P&L tracking	Calculate simulated P&L on every tick — mirror real exchange behaviour
Stop loss simulation	Trigger simulated stop when price hits $1.5 \times \text{ATR}$ from entry
Take profit simulation	Trigger simulated TP when RSI crosses back to 45-50 zone
Daily target tracking	Track simulated daily P&L — flag when $+0.5\%$ target hit
Trade logging	Log every paper trade: entry, exit, score, confirmations, session, duration, P&L
2-week paper trading run	Run the system for 2 weeks on live data with paper trades — validate results

MILESTONE 2 weeks of paper trading show consistent positive P&L with $< 5\%$ drawdown

Phase 4: Risk Cage — Risk Agent

Week 6-8

The Risk Agent gets built and tested. This is the guardian. Every risk rule is implemented, tested against historical worst-case scenarios, and validated. Nothing goes live without the Risk Agent being bulletproof.

Task	Details
Per-trade risk cap	Enforce max 0.25% of account per trade — verify Position Agent complies
Max concurrent positions	Block 4th trade regardless of score — test with rapid signals
Correlation blocker	If already long BTC, block ETH/SOL if correlation > 0.85
Losing streak breaker	3 consecutive losses → pause 30 min. 5 consecutive → stop for session
Daily loss limit	-1.5% → all trading stops. Test by simulating bad day.
Daily target shutoff	+0.5% → all trading stops. Verify no new trades after target hit.
Max drawdown kill switch	-8% from peak → everything stops. CEO must manually restart.
Session threshold enforcement	Asia: 5/5 only. London: 4/5. Prime: 3/5. Verify correct per session.
Stress testing	Run Risk Agent against Luna crash data, FTX collapse data, COVID crash data
Survival validation	Confirm the system survives worst historical scenarios without breaching limits
First real money test	Deploy with \$50-100 real capital. Tiny size. Validate real exchange execution.

MILESTONE Risk cage survives Luna/FTX/COVID stress tests. First real trade executes.

Phase 5: Context Layer — Research Agent

Week 8-11

The LLM-powered Research Agent adds contextual awareness. The system can now understand WHY price is dropping, not just that it is dropping. This is the difference between catching a normal dip and walking into a systemic collapse.

Task	Details
CryptoPanic API integration	Connect to news feed, receive structured articles every 5 minutes
Fear & Greed Index	Pull daily index from Alternative.me — add to Ontology
Claude API integration	Set up Anthropic SDK, design prompt for sentiment classification
Sentiment scoring pipeline	News article → Claude → bullish/bearish/neutral + score -1.0 to +1.0
Event type classification	Normal volatility vs systemic crisis vs regulatory vs exchange-specific
Confidence multiplier logic	Map sentiment score to multiplier (0.15x for crisis, 1.1x for positive)
ChromaDB setup	Vector database to store past news + outcomes for similarity search
News embedding pipeline	Convert articles to embeddings, store in ChromaDB with market reaction data
Historical calibration	Run past news through the pipeline, compare multiplier vs actual outcome
A/B test	Compare math-only performance vs math+context for 2 weeks — measure improvement
Telegram alerts	Send trade signals to your Telegram when system enters/exits positions

MILESTONE A/B test shows math+context outperforms math-only. Research Agent improves win rate.

By now the system has weeks of trade data. The Analytics Agent analyses everything and trains ML models to optimise the strategy. Fixed thresholds get replaced by learned weights. The system evolves from rule-following to pattern-recognising.

Analytics Agent build

Task	Details
Trade log database	Structured log: entry, exit, pair, TF, side, score, confirmations, session, P&L
Performance calculator	Rolling win rate, Sharpe, profit factor, expectancy — updated after each trade
Session breakdown	Which session produces best returns? Best win rate? Best risk-adjusted return?
Indicator contribution	Which confirmation check has highest predictive value? Rank them.
Daily report generator	Auto-generate PDF/HTML report every morning for CEO review
Dashboard API	FastAPI endpoints serving all metrics to the Next.js frontend

ML model training

Model	Purpose	Library
Random Forest classifier	Which indicator combos best predict bounces	scikit-learn
XGBoost classifier	Higher accuracy signal classification	xgboost
Logistic regression	Simple baseline to compare against	scikit-learn
Regime detection (HMM)	Classify market as normal/trending/volatile/crisis	hmmlearn
Indicator weight optimiser	Learn optimal weight for each confirmation check	scikit-learn
Sentiment calibration	Optimise the news multiplier based on outcomes	scikit-learn
Strategy decay detector	Alert if win rate trends downward over time	statsmodels
Position size optimiser	Learn optimal size given volatility and score	scikit-learn
Hyperparameter tuning	Automatically find best model settings	optuna

MILESTONE ML-optimised weights outperform fixed thresholds in backtest. Models deployed.

Phase 7: Full Deployment — All Systems Live

Week 14-16

Everything comes together. All 7 agents are live. ML models are deployed. The system trades autonomously with real capital. The CEO monitors and adjusts.

Task	Details
Capital deployment	Start with \$500-1,000. Scale only after 30 days of profitability.
Next.js dashboard	Build the CEO dashboard — live P&L, positions, metrics, risk gauges
Telegram notifications	Trade entries, exits, daily summary, risk alerts, kill switch notifications
Weekly model retrain	ML models retrain every Sunday on latest trade data
Monthly strategy review	CEO reviews monthly performance, adjusts targets, tweaks parameters
Compounding activation	Position sizes auto-scale with growing account balance
Daily target scaling	Month 1: 0.5%. Month 2-3: 0.75%. Month 4+: 1.0%. Based on proven consistency.
Documentation	All parameters, model versions, and changes documented in git
Redundancy	System running on both local machine and VPS backup
Scale capital	After 3 months profitability: deposit additional capital, increase position sizes

MILESTONE System is autonomous, profitable, and compounding. The AI hedge fund is live.

Complete Install & Setup Checklist

Everything you need to install before writing a single line of code.

Category	Package / Service	Install Command or Action
Python	Python 3.11+	brew install python (Mac) / python.org (Windows)
Exchange	ccxt	pip install ccxt
Indicators	pandas-ta	pip install pandas-ta
Data	pandas + numpy	pip install pandas numpy
Data	scipy	pip install scipy
Streaming	websockets + aiohttp	pip install websockets aiohttp
Database	PostgreSQL	brew install postgresql / apt install postgresql
Database	psycpg2	pip install psycpg2-binary
Cache	Redis	brew install redis / apt install redis
Cache	redis-py	pip install redis
API	FastAPI + uvicorn	pip install fastapi uvicorn
LLM	anthropic SDK	pip install anthropic
Vector DB	chromadb	pip install chromadb
Embeddings	sentence-transformers	pip install sentence-transformers
ML	scikit-learn	pip install scikit-learn
ML	xgboost	pip install xgboost
ML	hmmlearn	pip install hmmlearn
ML	optuna	pip install optuna
ML	statsmodels	pip install statsmodels
Dashboard	Node.js + Next.js	brew install node && npx create-next-app
Charts	plotly	pip install plotly
Notifications	python-telegram-bot	pip install python-telegram-bot
Account	MEXC API keys	MEXC → Account → API Management → Create key
Account	Bybit API keys	Bybit → Account → API Management → Create key
Account	Anthropic API key	console.anthropic.com → API Keys → Create
Account	CryptoPanic API key	cryptopanic.com → API → Get free key
Account	Telegram Bot	BotFather on Telegram → /newbot → save token

The rule: complete each phase's milestone before starting the next. No skipping. No shortcuts. Each phase produces a working, testable deliverable.